

Photoplethysmography (PPG) measures the heart rate based on the absorption of visible light in the body. An LED illuminates the tissue and a light detector monitors absorption, as shown in Figure 1. The tissue blood volume (B_V) has a large influence on the light absorption. B_V increases when the heart contracts and decreases when the heart relaxes, causing the PPG signal to increase and decrease in response to the changes in B_V .

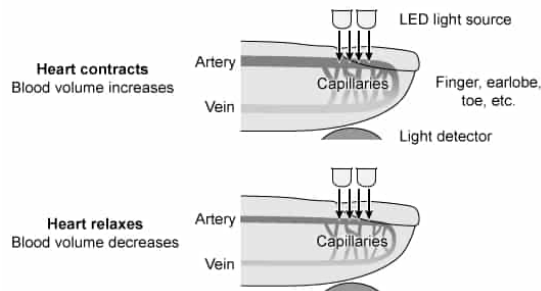


Figure 1 Diagram of a PPG system during the cycle of heart contractions and relaxations

The LED circuit consists of a voltage source, a current-limiting resistor, and the LED connected in series. The LED itself does not behave like a resistor: When current flows through the LED, the LED voltage remains constant, known as the forward voltage. The LED cannot handle large currents, so the current-limiting resistor keeps the current in the circuit below the maximum forward current of the LED.

The PPG signal shows when blood flows into the tissue, but an ECG indicates heart contractions. Figure 2 compares these signals, revealing a time delay that depends on the pulse wave velocity, which is the rate the pressure wave moves through the vasculature. Healthy blood vessels dilate when blood pressure increases, slowing the pulse wave as it leaves the heart. Increasing age and some diseases cause "hardening" of the blood vessels, increasing the pulse wave velocity.

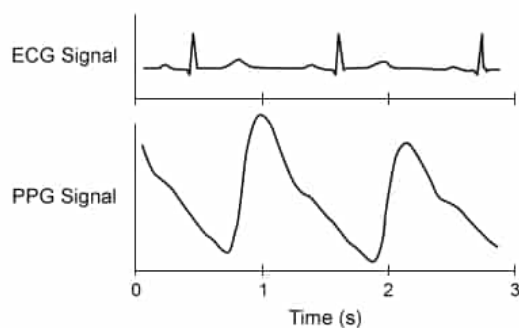


Figure 2 Simultaneous ECG signal and PPG signal showing the effects of the pulse wave velocity

What information about the LED circuit is required to calculate which resistance value should be used?

- ☐ A. Source voltage and LED maximum forward current [49%]
- ☐ B. Wavelength of light, LED forward voltage, and LED maximum forward current [2%]

- ✓ ☒ C. Source voltage, LED forward voltage, and LED maximum forward current [23%]
☐ D. LED forward voltage and LED maximum forward current [23%]

Correct

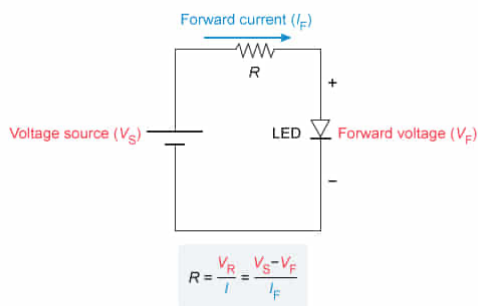
24% answered correctly

Explanation:

Applying Ohm's law to calculate resistance for LED circuit

Ohm's law $V_R = IR$

- V_R Voltage across R
- I Current through R
- R Resistance



R depends on source voltage, LED forward voltage, and LED maximum forward current.

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Ohm's law states that the **voltage** V across a resistor equals the **current** I through the resistor multiplied by the **resistance** R :

$$V = IR$$

Ohm's law can be rearranged to calculate the resistance as the ratio of V and I :

$$R = \frac{V}{I}$$

Consequently, the value of the resistance depends only on the voltage across the resistor and the current through the resistor.

In this question, the resistor and the LED are connected in a **series circuit**, so the same current flows through each element. Thus, the LED maximum forward current I_F is the same as the current through the resistor:

$$I = I_F$$

By conservation of energy, the voltage rise from the voltage source V_S is equal to the sum of the voltage drops across the other elements in the circuit, the resistor V_R and the LED V_F :

$$V_S = V_R + V_F$$

Consequently,

$$V_R = V_S - V_F$$

and Ohm's law implies that

$$R = \frac{V_S - V_F}{I_F}$$

Therefore, the two voltages and the current are needed to calculate the resistance.

(Choices A and D) Ohm's law implies that neither the source voltage nor the LED voltage are sufficient by themselves to calculate the voltage across the resistor.

(Choice B) The wavelength of the light does not affect the resistor value, and the LED voltage alone is not sufficient to calculate the voltage across the resistor.

Educational objective:

In a series circuit, the same current flows through each circuit component. Resistance can be calculated as the ratio of voltage to current using Ohm's law.

Time spent:0 sec

Subject:
Physics

QID:403166

System:
Electrostatics and Circuits

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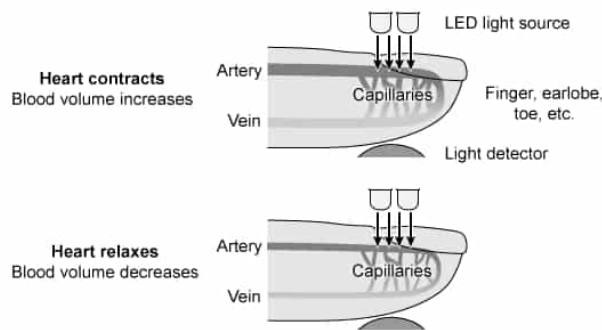


Figure 1 Diagram of a PPG system during the cycle of heart contractions and relaxations

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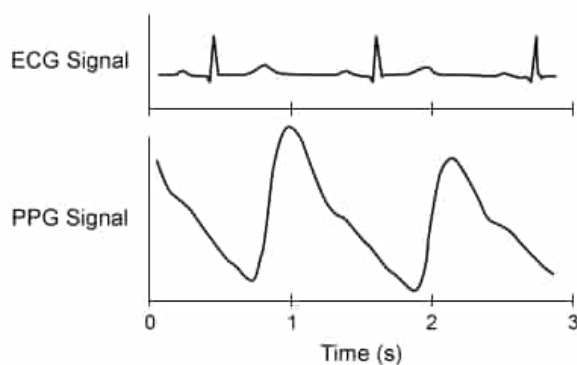


Figure 2 Simultaneous ECG signal and PPG signal showing the effects of the pulse wave velocity

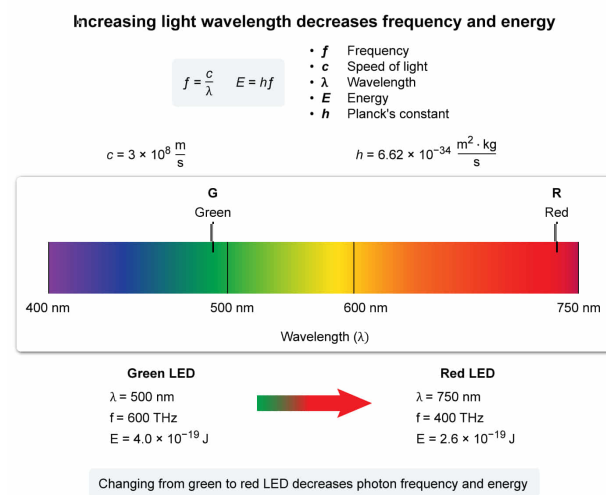
If the light source for the PPG system is changed from a green LED to a red LED, what would be the effect on the frequency and the energy of the emitted photons?

- ☐ A. The frequency and energy would increase. [21%]
- ✓ ☒ B. The frequency and energy would decrease. [69%]
- ☐ C. The frequency would increase, and the energy would decrease. [6%]
- ☐ D. The frequency and energy would not change. [1%]

Correct

70% answered correctly

Explanation:



The **electromagnetic spectrum** shows that different colors of visible light have different **wavelengths** (λ). For example, the wavelength of green light is around 500 nm and the wavelength of red light is around 750 nm. The **frequency** f of light is equal to the speed of light c divided by its wavelength λ :

$$f = \frac{c}{\lambda}$$

Furthermore, the **energy** E of the light's photons is equal to the product of Planck's constant h and f :

$$E = hf$$

Substituting the equation for frequency yields:

$$E = \frac{hc}{\lambda}$$

Hence, both the frequency and the photon energy are **inversely proportional** to the wavelength of light.

In this question, changing from a green LED to a red LED increases the wavelength of the light. Therefore, increasing the wavelength would cause both the frequency and the energy of the photons to decrease.

(Choice A) The frequency and energy would decrease, not increase, as the wavelength increases.

(Choice C) The energy would decrease as the wavelength increases, but the frequency would also decrease.

(Choice D) The frequency and energy would decrease, not stay the same, as the wavelength increases.

Educational objective:

The wavelength of light is inversely proportional to its frequency and energy. Increasing the wavelength decreases the frequency and energy.

Time spent:0 sec

Subject:
Physics

QID:403167

System:
Electrostatics and Circuits

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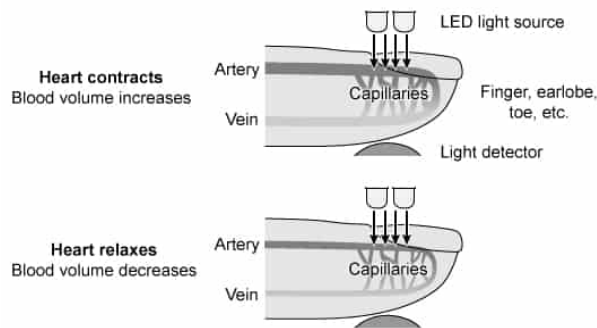


Figure 1 Diagram of a PPG system during the cycle of heart contractions and relaxations

The LED circuit consists of a voltage source, a current-limiting resistor, and the LED connected in series. The LED itself does not behave like a resistor: When current flows through the LED, the LED voltage remains constant, known as the forward voltage. The LED cannot handle large currents, so the current-limiting resistor keeps the current in the circuit below the maximum forward current of the LED.

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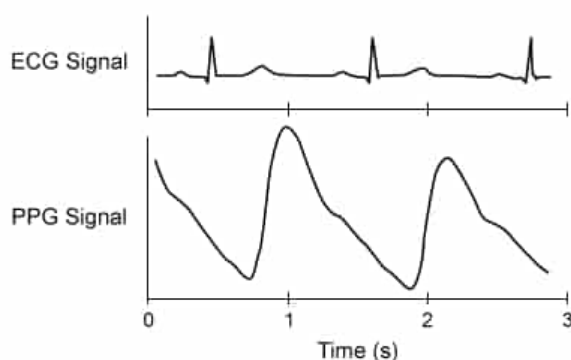


Figure 2 Simultaneous ECG signal and PPG signal showing the effects of the pulse wave velocity

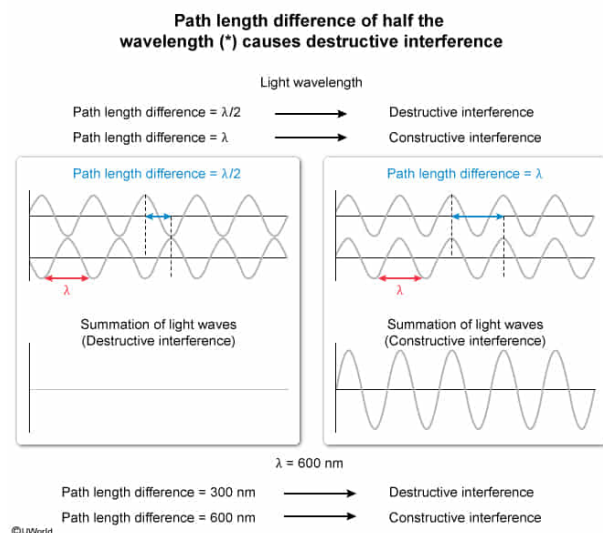
Light can be scattered as it travels through tissue, resulting in different path lengths from the LED to the detector. The LED generates light at 600 nm. This wavelength leads to destructive interference when the path lengths differ by:

- ☐ A. 150 nm [10%]
- ✓ ☒ B. 300 nm [60%]
- ☐ C. 600 nm [21%]
- ☐ D. 1,200 nm [7%]

Correct

60% answered correctly

Explanation:



When two waves meet at the same point, they overlap with one another and cause **wave interference**. Pure constructive interference occurs when the peaks and troughs of the two waves overlap exactly, meaning the **phase difference** between the waves is 0° . This corresponds to a path length difference Δd between the two waves of zero or an integer multiple of the wavelength λ :

$$\Delta d = 0, \lambda, 2\lambda, 3\lambda, \dots$$

When the peaks of one wave overlap exactly with the troughs of the other wave, the phase difference between the waves is 180° and pure **destructive interference** occurs. This happens when the difference in path length is half the wavelength or an odd multiple of half the wavelength:

$$\Delta d = \frac{\lambda}{2}, \frac{3\lambda}{2}, \dots$$

In this question, the wavelength is 600 nm and destructive interference occurs when the path length difference is 300 nm, 900 nm, 1,500 nm, and so on. When the

difference in the path lengths is 300 nm, the sum of the two waves yields a zero signal consistent with destructive interference.

(Choice A) A difference in path length of 150 nm is one fourth of the wavelength, yielding a phase shift of 90° . Summing two waves shifted by 90° results in a combination of both constructive and destructive interference.

(Choices C and D) Differences in path length of 600 nm or 1,200 nm are multiples of the wavelength, yielding a phase shift of 360° and pure constructive interference.

Educational objective:

Destructive interference occurs when the path lengths of two waves differ by half the wavelength. Differences in path length that are integer multiples of the wavelength cause constructive interference.

Time spent:0 sec

Subject:

Physics

QID:403168

System:

Electrostatics and Circuits

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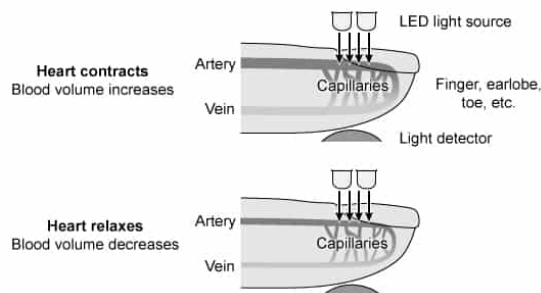


Figure 1 Diagram of a PPG system during the cycle of heart contractions and relaxations

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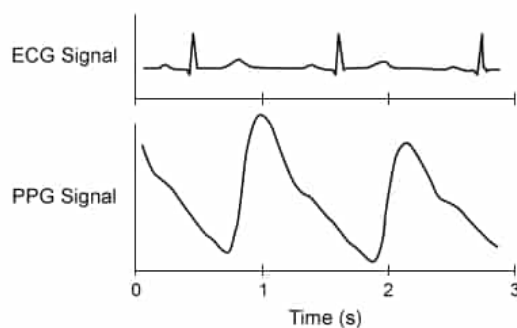


Figure 2 Simultaneous ECG signal and PPG signal showing the effects of the pulse wave velocity

The ECG and PPG signals show a time delay between the heart contraction and the inflow of blood into the finger due to the pulse wave velocity. What other information is needed to calculate the average pulse wave velocity?

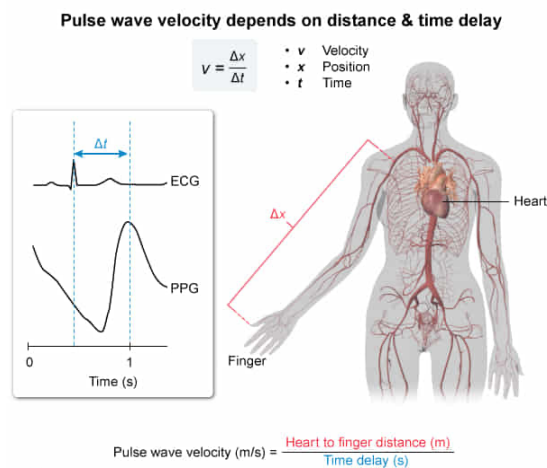
- ✓ ☒ A. Distance from the heart to the finger [67%]
- ☐ B. Average blood pressure [16%]

- ☐ C. Relative blood volume in the finger [5%]
- ☐ D. Average acceleration of the blood [11%]

Correct

66% answered correctly

Explanation:



Calculating the average **velocity** v requires knowing the change in position Δx over an interval Δt :

$$v = \frac{\Delta x}{\Delta t}$$

The change in position is calculated from the starting position x_0 and the final position x_f :

$$\Delta x = x_f - x_0$$

Similarly, the interval is calculated from the starting time t_0 and the ending time t_f :

$$\Delta t = t_f - t_0$$

In this question, the starting position of the pulse wave is the heart, and the final position is the finger where the PPG signal is collected. The starting time of the pulse wave is the large spike in the **ECG signal**, which corresponds to the contraction of the heart. The ending time is the peak in the PPG signal, indicating arrival of the pulse wave in the finger.

The average pulse wave velocity v_{PW} is calculated as the ratio between the distance from the heart to the finger and the time delay between the ECG peak and the PPG peak:

$$v_{PW} = \frac{\text{heart to finger distance}}{\text{time delay}}$$

Therefore, the only information needed to calculate the average pulse wave velocity is the distance from the heart to the finger and the time delay.

(Choices B, C and D) Although these parameters may influence the velocity of the blood itself, they are not necessary to find the pulse wave velocity if the time needed for the wave to move between two points is known. The only additional information needed for this calculation

is the distance between the two points.

Educational objective:

The velocity of the pulse wave is calculated by dividing the change in position by the time interval. The pulse wave travels from the heart to the finger during the delay between the peak in the ECG signal and the peak in the PPG signal.

Time spent:0 sec

Subject:
Physics

QID:403169

System:
Electrostatics and Circuits

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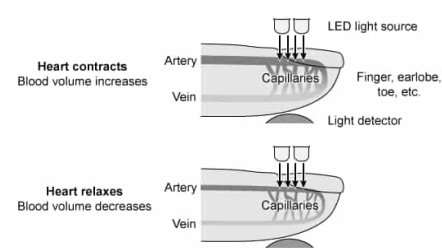


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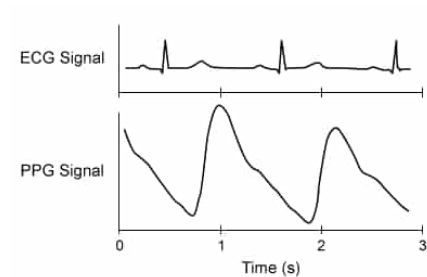
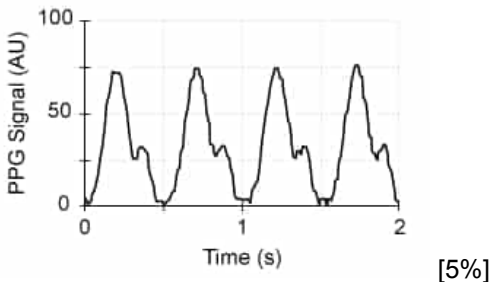


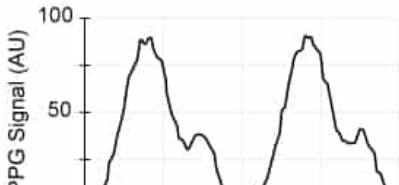
Figure 2 Simultaneous ECG signal and PPG signal showing the effects of the pulse wave velocity

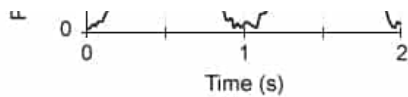
Which of the PPG signals shown below represents a heart rate of 90 beats per minute (bpm)?

☐ A.



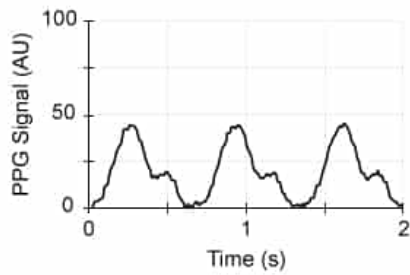
☐ B.





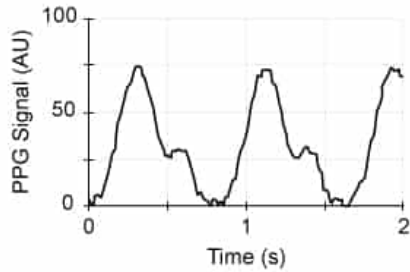
[19%]

✓ ☒ C.



[60%]

☐ D.



[13%]

Correct

61% answered correctly

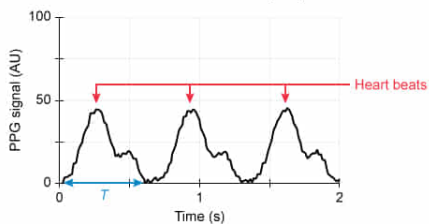
Explanation:

Calculation of heart rate

$$f = \frac{1}{T}$$

- f Frequency
- T Period

$$\text{Heart Rate (bpm)} = \frac{1}{T (\text{min})}$$



$$\text{Heart rate} = \frac{1}{0.66 \text{ s}} = 1.5 \frac{\text{beats}}{\text{s}} \times \frac{60 \text{ s}}{1 \text{ min}} = 90 \frac{\text{beats}}{\text{min}}$$

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A periodic signal repeats itself over and over. The **frequency** f of the signal is the number of times it repeats over a specific interval, and the **period** T is the time required to complete one cycle of the signal. The frequency is equal to the inverse of the period:

$$f = \frac{1}{T}$$

In this question, the **heart rate** HR corresponds to the frequency of the heartbeat and can be calculated by estimating the period of one beat for each answer choice. HR is calculated in beats per minute (bpm), so T is converted from seconds to minutes

Therefore, the graph shown in Choice C, where each heartbeat has a period of 0.66 seconds, corresponds to 90 bpm:

$$HR = \frac{1}{0.66 \text{ s}} \times \frac{60 \text{ s}}{1 \text{ min}} = 90 \frac{\text{beats}}{\text{min}}$$

(Choice A) This plot shows each heartbeat has a period of 0.5 s, giving a **heart rate of 120 bpm**.

(Choice B) This plot shows each heartbeat has a period of 1 s, giving a heart rate of 60 bpm.

(Choice D) This plot shows each heartbeat has a period of 0.8 s, giving a heart rate of 75 bpm.

Educational objective:

The PPG signal is periodic with a frequency that corresponds to the heart rate. The heart rate can be calculated from a plot of the PPG signal by determining the period and calculating its inverse.

Time spent:0 sec

Subject:
Physics

QID:403170

System:
Electrostatics and Circuits